

1 QC Results From DNA Extraction

Sample ID	Nucleic Acid (ng/ul)	A260 (Abs)	A280 (Abs)	260/280
C4	1.0	0.019	0.011	1.70
C5	0.9	0.017	0.011	1.63
C12	2.1	0.042	0.009	1.75

Table 1: The QC results from my group's DNA extraction samples.

2 Gel Runs

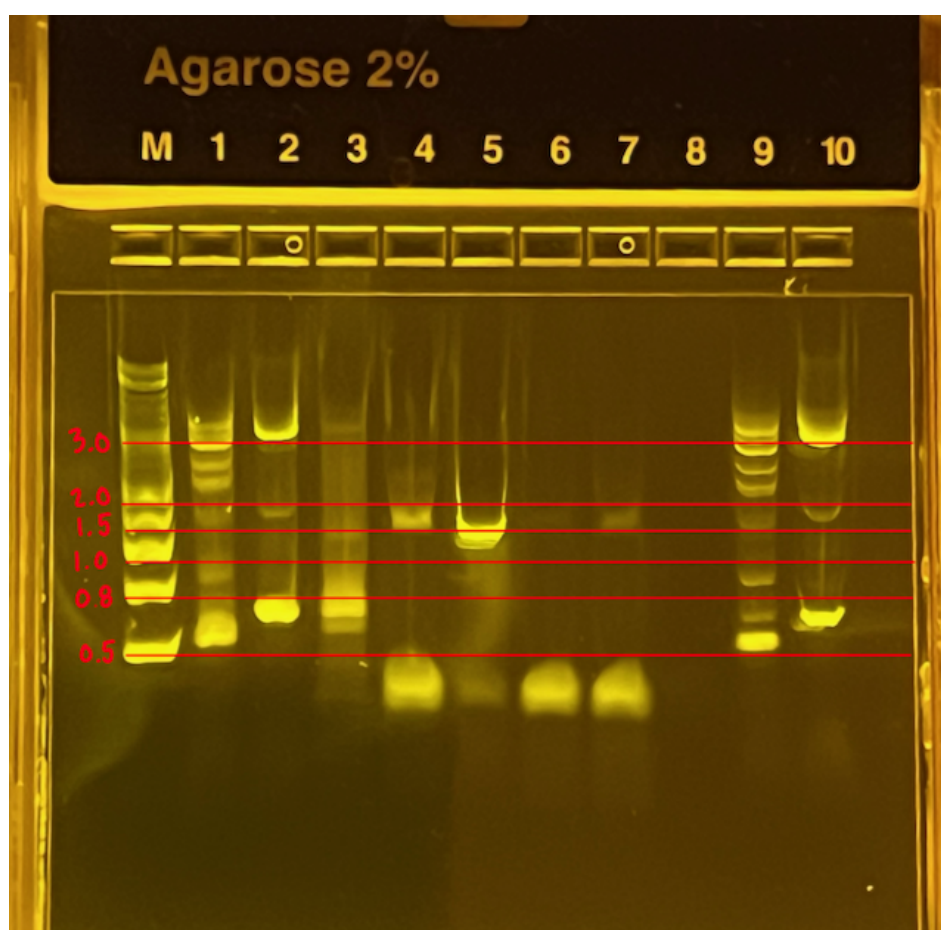


Figure 1: Our gel electrophoresis run. Number of bases marked along the ladder along lane M in red in kilobases. Contents of each lane are in the table below.

In lanes 1 and 2 (and their duplicates in lanes 9 and 10) we see good products that we wanted. This suggests that the primers we created did create a product in these samples. However, with the primer we added, we would expect a product of about 980 bases, which is not what we see here (it looks to be about 600 – 700 bases). This might be due to the primers bonding in incorrect places, or perhaps the sampled DNA is not one of the six that we tested upon.

In lane 3, we see a vague response at about 800 bases as well. This is closer to what we wanted of 980 bases. If this is the intended output that we wanted, this implies sample C12 had A1, A5, A6, or A7 within the sample

Lane Number	DNA	Primer
1	C4	Our Primer
2	C5	Our Primer
3	C12	Our Primer
4	C4	Kangas Primer
5	C5	Kangas Primer
6	C12	Kangas Primer
7	None	Our Primer (Primer Control)
8	C4, C5, C12	None (DNA Control)
9	C4	Our Primer
10	C5	Our Primer

Table 2: The contents of each lane in the gel electrophoresis run.

(via our PCR lab).

Lanes 4 and 5 had good responses with the Kangas primer. This is good since it implies that there was indeed the correct type of DNA within the C4 and C5 samples.

Lane 6 doesn't seem to have much other than many really short DNA strands, which might be the primers. This somewhat weakens the idea that lane 3 had a true response, since if it did this lane would also be likely to have a response.

Lanes 7 and 8 are our controls and work as we would expect them to. We see the primers in the bottom of lane 7, and we see nothing in lane 8 since the DNA should be longer than anything on this scale.

3 DNA Identification

I looked specifically at the C5 sample. This DNA sample was IDed as the haemolytic strain of *Staphylococcus*. My research led me to find that this is part of the "skin flora" of humans. I collected this sample from the handle of a doorway into a section of Wean Hall. Thus, having a strain of bacteria from human skins makes a lot of sense.

4 Conclusions

In conclusion, I believe we did quite well with extracting the DNA. Our sample was pure enough that the identification company found a strong match for the bacteria. Additionally, we had PCR results for many of our extractions, which means that they did indeed have DNA in them.

On the other hand, our primers did not quite work as intended. As mentioned earlier the length of the products were not as expected. This error could have been caused by incorrect selection of primers, a poor gel run which skewed the read lengths, or a corruption of the DNA contents, which caused the primers to react with some other DNA that made products of the length observed.